

Exercise 04

```
In [2]: import os
import openai
import tiktoken
from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file

openai.api_key = os.environ['OPENAI_API_KEY']
```

helper function

This may look familiar if you took the earlier course "ChatGPT Prompt Engineering for Developers" Course.

Throughout this course, we will use OpenAI's `gpt-3.5-turbo` model and the [chat completions endpoint](#).

This helper function will make it easier to use prompts and look at the generated outputs.

Note: In June 2023, OpenAI updated gpt-3.5-turbo. The results you see in the notebook may be slightly different than those in the video. Some of the prompts have also been slightly modified to produce the desired results.

```
In [4]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in physics."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "Explain how black holes are formed."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Black holes are formed when a massive star reaches the end of its life cycle and undergoes a supernova explosion. During the supernova, the outer layers of the star are blasted away into space, leaving behind the core of the star, which collapses under its own gravity.

If the core of the star is more massive than about 3 times the mass of the Sun, the gravitational forces are strong enough to overcome all other forces, resulting in the formation of a black hole. The core collapses into a singularity, a point of infinite density and zero volume, surrounded by an event horizon from which not even light can escape.

This process transforms the massive star into a region of spacetime where gravity is so intense that the laws of physics as we know them break down. The formation of a black hole is a fascinating and extreme consequence of the fundamental principles of general relativity.

```
In [17]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are an expert in event planning."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "I need to plan a birthday party for my mother."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

I'd be happy to help you with that! Here are some steps to consider when planning a birthday party for your mother:

1. ****Set a Budget****: Determine how much you want to spend on the party. This will help you make decisions about the venue, food, decorations, and entertainment.
 2. ****Choose a Date and Time****: Check with your mother to see when she is available and choose a date and time that works for her.
 3. ****Select a Venue****: Decide whether you want to have the party at home, at a restaurant, or at a special event space. Consider the number of guests and the type of atmosphere your mother would enjoy.
 4. ****Create a Guest List****: Make a list of everyone your mother would like to invite. Consider friends, family members, and any of her favorite people.
 5. ****Send Invitations****: Whether you choose to send physical invites or create a digital invitation, make sure to include important details like the date, time, location, and RSVP information.
 6. ****Plan the Menu****: Decide what food and drinks you want to serve at the party. Consider your mother's favorite dishes and any dietary restrictions of your guests.
 7. ****Decorate the Space****: Think about your mother's favorite colors and themes when decorating the party space. You can incorporate balloons, banners, flowers, and other decorations to create a festive atmosphere.
 8. ****Plan Entertainment****: Consider what activities or entertainment your mother would enjoy. This could include music, games, a photo booth, or a special performance.
 9. ****Organize a Special Tribute****: Consider planning a special tribute to your mother, such as a slideshow of her life or a speech expressing your love and appreciation.
 10. ****Capture Memories****: Make sure to have a photographer or designate someone to take photos and videos throughout the party to capture the special moments.
 11. ****Thank You Cards****: After the party, don't forget to send thank you cards to your guests for attending and to show your appreciation for their presence.
- Remember, the most important thing is to make sure your mother feels loved and celebrated on her special day. Have fun planning the party!

```
In [19]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You're an expert in Entertainment field."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "Tell me what you know about Jennifer Lawrence."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Jennifer Lawrence is an American actress known for her versatile roles in both independent and big-budget films. She gained widespread recognition for her performances in the "The Hunger Games" film series, where she portrayed the character Katniss Everdeen. Lawrence also won critical acclaim and multiple awards, including an Academy Award for Best Actress, for her roles in films such as "Silver Linings Playbook" and "American Hustle." She is known for her talent, down-to-earth personality, and sense of humor, making her one of the most popular and highest-paid actresses in Hollywood.